IGCSE 0607 Practice Questions — E9.3 — Combined Questions & Markscheme

Probability of combined events

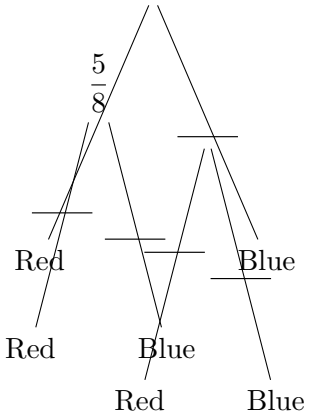
Total Questions: 39 — **Total Marks:** 265 Marks

Table of Contents & Quick Navigation

- **E9.3** Probability of combined events
  → **Paper 2 (Non-Calculator)** *(18 Qs)* · **Paper 4 (Calculator)** *(21 Qs)*

1 A bag contains 5 red counters and 3 blue counters.
A counter is taken at random from the bag, its colour is noted and it is replaced.
A second counter is then taken at random.

(a) Complete the tree diagram.



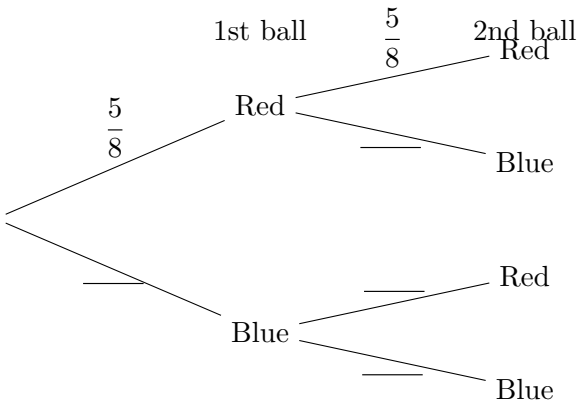
[2]

(b) Calculate the probability that both counters are red.

..... **[2]**

2 A bag contains 8 balls.
5 of the balls are Red and 3 of the balls are Blue.
A ball is taken at random from the bag, its colour is noted, and it is replaced.
A second ball is then taken at random from the bag.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that both balls are Red.

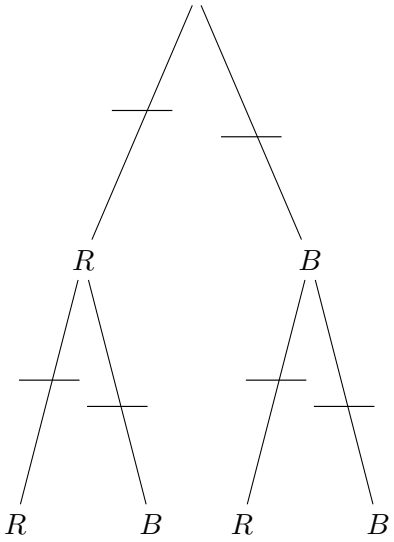
..... **[1]**

(c) Calculate the probability of picking one Red ball and one Blue ball in any order.

..... **[2]**

3 A bag contains 7 red counters and 5 blue counters.
A counter is taken at random, its colour is noted, and it is NOT replaced in the bag.
A second counter is then taken at random and its colour is noted.

(a) Complete the tree diagram by writing the missing probabilities on the branches.



[3]

(b) Calculate the probability that both counters are red.

..... **[2]**

(c) Calculate the probability that one counter is red and one counter is blue.

..... **[2]**

4 A box contains x red balls and 4 blue balls.
Two balls are taken from the box at random, without replacement.
The probability that both balls are red is $\frac{1}{3}$.

(a) Show that $x^2 - 5x - 6 = 0$.

[3]

(b) Work out the number of red balls in the box.

..... **[2]**

(c) Calculate the probability that the two balls taken have different colours.

..... **[3]**

5 A box contains n counters.
3 of the counters are Red and all the remaining counters are Blue.
Two counters are taken at random from the box, without replacement.
The probability that both counters are Red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the value of n.

$n =$ ..... **[2]**

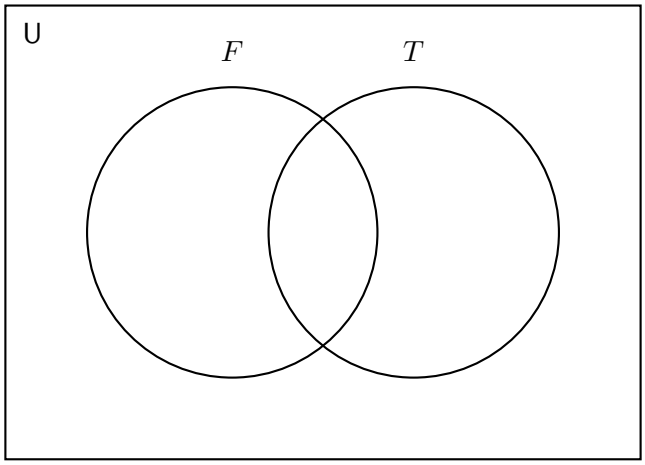
(c) Calculate the probability that exactly one of the two counters chosen is Red.

..... **[3]**

6 There are 30 students in a class.
U = {students in the class}
F = {students who play Football}
T = {students who play Tennis

18 students play Football. 14 students play Tennis. 4 students do not play Football and do not play Tennis.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

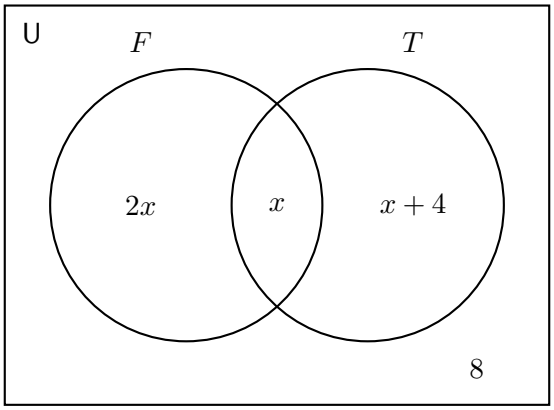
(b) A student is chosen at random from the class.
Find $P(F \cap T)$.

$P(F \cap T) =$ ..... **[1]**

(c) Two students are chosen at random from the class.
Calculate the probability that both students play Football but not Tennis.

..... **[3]**

7 A group of 40 students are asked if they play Football (F) or Tennis (T).
The Venn diagram shows the number of students in each region in terms of x.



(a) Find the value of x.

$x =$ ..... **[2]**

(b) Find $P(F \cup T)$.

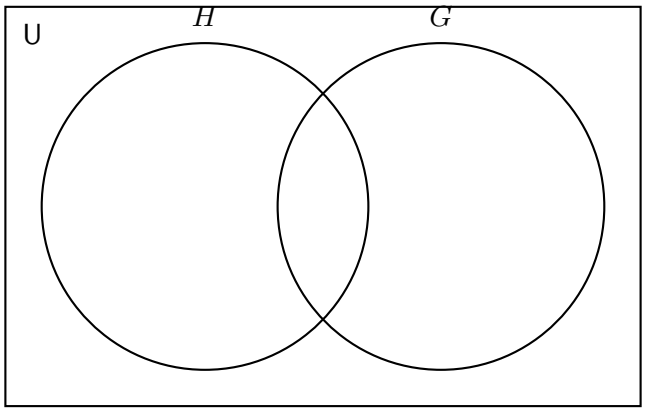
..... **[1]**

(c) Two students are chosen at random from the group of 40.
Calculate the probability that both students play Football only.

..... **[3]**

8 In a group of 50 students, 32 study History (H), 21 study Geography (G), and 6 study neither subject.

(a) Complete the Venn diagram to show the number of students in each region.



[3]

(b) A student is chosen at random from the group of 50.

Find:

(i) $P(H \cap G)$

$P(H \cap G) =$ ..... **[1]**

(ii) $P(H \cup G')$

$P(H \cup G') =$ ..... **[1]**

(c) Two students who study History are chosen at random.
Calculate the probability that both students also study Geography.

..... **[2]**

9 A bag contains 4 red discs and some blue discs.
Let the total number of discs in the bag be x.
Two discs are taken at random from the bag, without replacement.
The probability that both discs are red is $\frac{1}{11}$.

(a) Show that $x^2 - x - 132 = 0$.

[3]

(b) Find the number of blue discs in the bag.

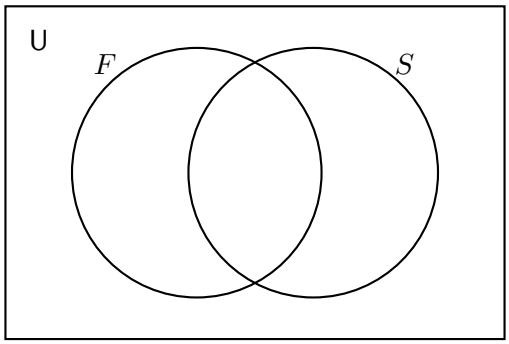
..... **[3]**

(c) Calculate the probability that one disc is red and one disc is blue.

..... **[3]**

10 A class of 30 students were surveyed about the subjects they study.
17 study French (F), 14 study Spanish (S), and 5 study neither subject.

(a) Complete the Venn diagram to represent this information.



[2]

(b) A student is chosen at random from the class.
Work out the probability that this student:

(i) studies both French and Spanish,

..... **[1]**

(ii) studies French but not Spanish,

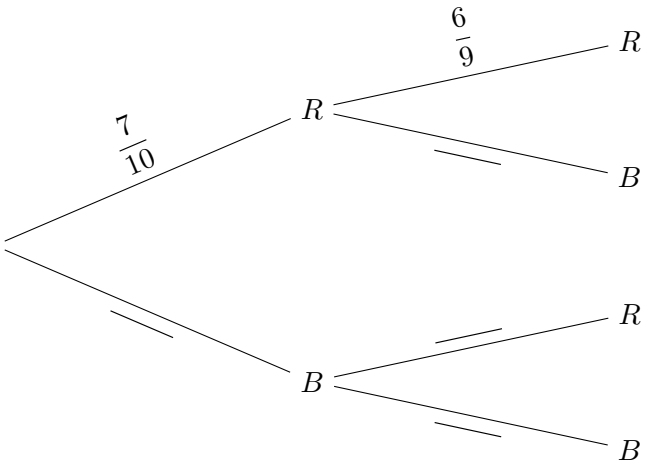
..... **[1]**

(iii) is a member of the set $F \cup S$.

..... **[1]**

11 A box contains 7 red counters and 3 blue counters.
A counter is chosen at random from the box and its colour is recorded.
It is not replaced.
A second counter is then chosen at random from the box.

(a) Complete the tree diagram.



[3]

(b) Calculate the probability that both counters chosen are red.

..... **[2]**

(c) Calculate the probability that the two counters chosen are of different colours.

..... **[2]**

12 A box contains x blue counters and 4 green counters.
Two counters are chosen at random from the box, one after another, without replacement.

(a) Show that the probability that both counters chosen are blue is $\dfrac{x(x-1)}{x^2+7x+12}$.

[2]

(b) The probability that both counters chosen are blue is $\dfrac{1}{3}$.

(i) Show that $x^2 - 5x - 6 = 0$.

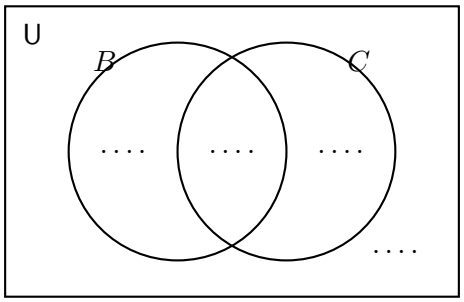
[3]

(ii) Solve the equation $x^2 - 5x - 6 = 0$ to find the number of blue counters in the box.

$x =$ ..... **[3]**

13 In a group of 30 students, 18 study Biology (B), 15 study Chemistry (C), and 4 study neither Biology nor Chemistry.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

(b) A student is chosen at random from the group.

(i) Find the probability that this student studies Chemistry but not Biology.

..... **[1]**

(ii) Find $\mathrm{P}(B \cup C)$.

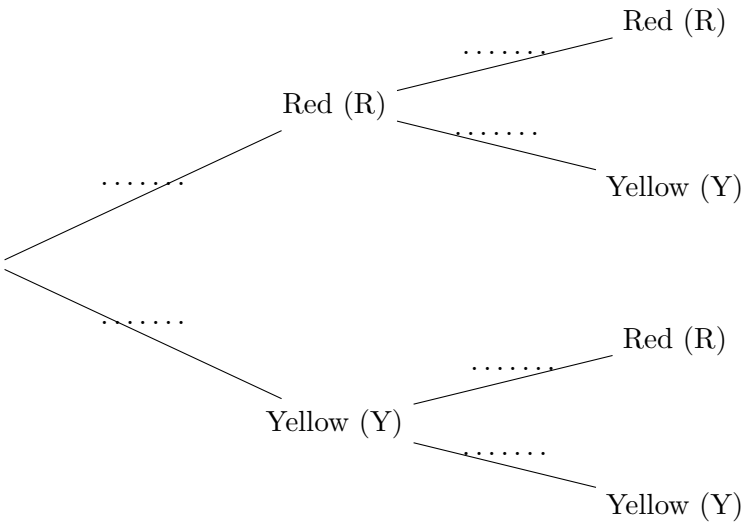
..... **[1]**

(iii) Given that the student chosen studies Biology, find the probability that this student also studies Chemistry.

..... **[1]**

14 A bag contains 5 red sweets and 4 yellow sweets.
Maya takes a sweet at random from the bag and eats it.
She then takes a second sweet at random from the bag and eats it.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that Maya eats two sweets of the same colour.

..... **[2]**

(c) Calculate the probability that Maya eats at least one yellow sweet.

..... **[3]**

15 Box A contains x blue counters and 3 red counters.
Box B contains 4 blue counters and 2 red counters.
A counter is taken at random from Box A and placed into Box B.
A counter is then taken at random from Box B.

(a) Show that the probability of taking a blue counter from Box B is

$$\frac{5x+12}{7(x+3)}$$

[4]

(b) The probability of taking a blue counter from Box B is $\frac{3}{4}$.

(i) Work out the value of x.

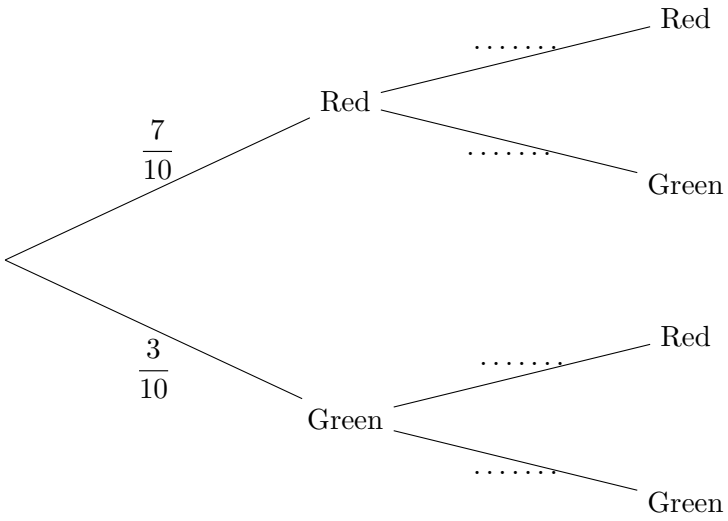
$x =$ ..... **[3]**

(ii) Using your value of x, find the probability that the counter transferred from Box A to Box B was red, given that the counter taken from Box B is blue.

..... **[1]**

16 A bag contains 7 red apples and 3 green apples.
Liam takes two apples at random, without replacement.

(a) Complete the tree diagram.



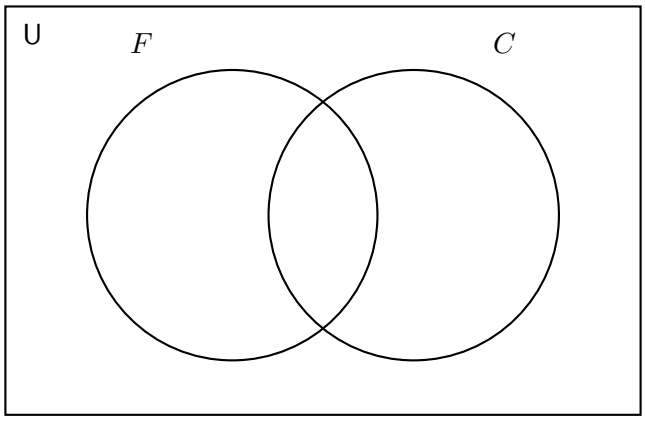
[2]

(b) Work out the probability that Liam takes two apples of the same colour.

..... **[3]**

17 In a group of 40 students, 24 like football (F) and 18 like cricket (C).
x students like both sports.
3 students do not like either sport.

(a) Complete the Venn diagram to show this information in terms of x.



[2]

(b) Work out the value of x.

$x =$ ..... **[2]**

(c) Two students are chosen at random from the group of 40.
Work out the probability that one student likes football only and the other student likes cricket only.

..... **[3]**

18 A bag contains n balls.
3 of the balls are red and the rest are blue.
Two balls are chosen at random, without replacement.
The probability that both balls are red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the value of n.

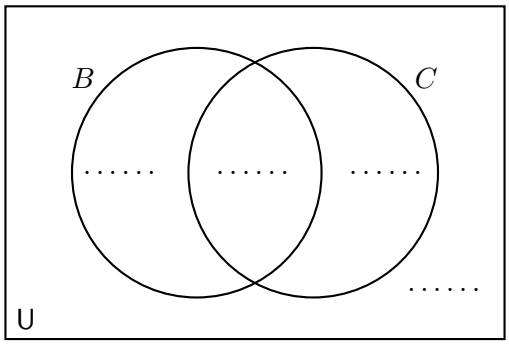
$n =$ ..... **[2]**

(c) Find the probability that the two balls have different colours.

..... **[3]**

19 In a group of 30 students, 18 study Biology (B) and 15 study Chemistry (C).
5 students study neither Biology nor Chemistry.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

(b) A student is chosen at random from the group.

(i) Find the probability that this student studies Biology but not Chemistry.

..... **[1]**

(ii) Find $\mathrm{P}(B \cup C)$.

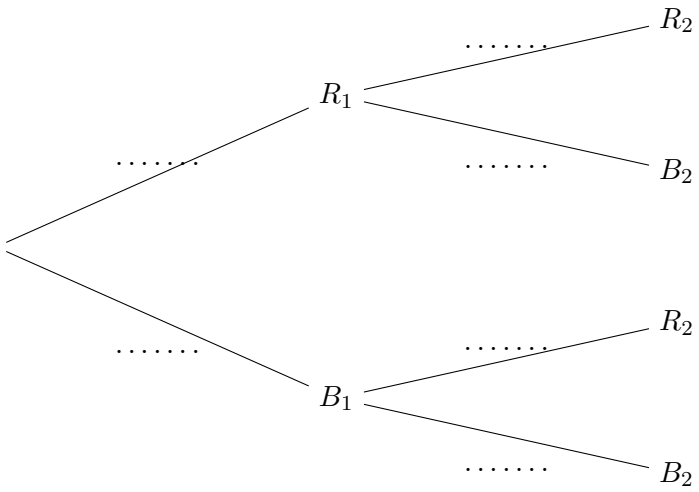
..... **[1]**

(c) Find the probability that a student chosen at random from those who study Chemistry also studies Biology.

..... **[2]**

20 A bag contains 5 red counters and 3 blue counters.
Two counters are taken from the bag at random, without replacement.

(a) Complete the tree diagram by writing the missing probabilities on the branches.



[3]

(b) Calculate the probability that both counters taken are the same colour.

..... **[2]**

(c) Calculate the probability that at least one of the counters taken is red.

..... **[2]**

21 A box contains n chocolates, of which 4 are dark chocolates and the rest are milk chocolates.

(a) Two chocolates are taken from the box at random, without replacement.
The probability that both chocolates are dark is $\frac{2}{15}$.
Show that $n^2 - n - 90 = 0$.

[3]

(b) Solve the equation $n^2 - n - 90 = 0$ to find the total number of chocolates in the box.

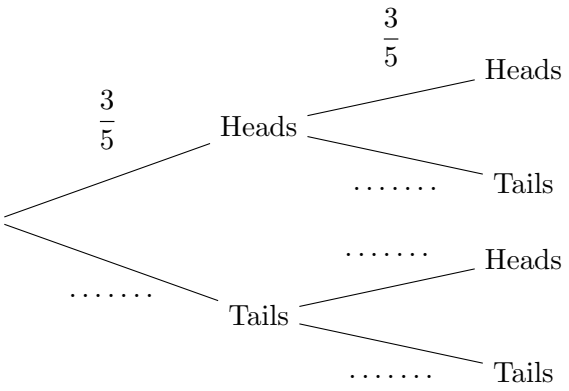
$n =$ ..... **[2]**

(c) Three chocolates are now taken at random from the original box of chocolates, without replacement.
Work out the probability that exactly two of the three chocolates taken are dark chocolates.

..... **[4]**

22 A biased coin has a probability of showing Heads of $\frac{3}{5}$.
The coin is flipped twice.

(a) Complete the tree diagram.



[2]

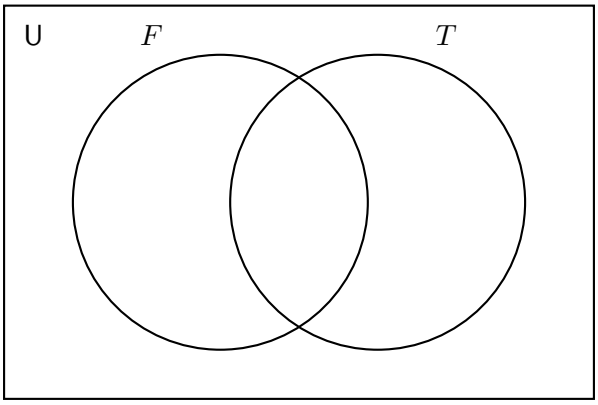
(b) Calculate the probability that the coin shows exactly one Head.

..... **[2]**

23 In a group of 40 students:

- 24 play Football (F)
- 19 play Tennis (T)
- 7 do not play either sport.

(a) Complete the Venn diagram to show this information.



[2]

(b) A student is chosen at random from the group.
Find the probability that the student plays both Football and Tennis.

..... **[1]**

(c) Two students who play Football are chosen at random.
Find the probability that exactly one of them also plays Tennis.

..... **[3]**

24 A bag contains n counters.
3 of the counters are red and the rest are blue.
Two counters are chosen at random, without replacement.
The probability that both counters are red is $\frac{1}{15}$.

(a) Show that $n^2 - n - 90 = 0$.

[3]

(b) Find the number of blue counters in the bag.

..... **[2]**

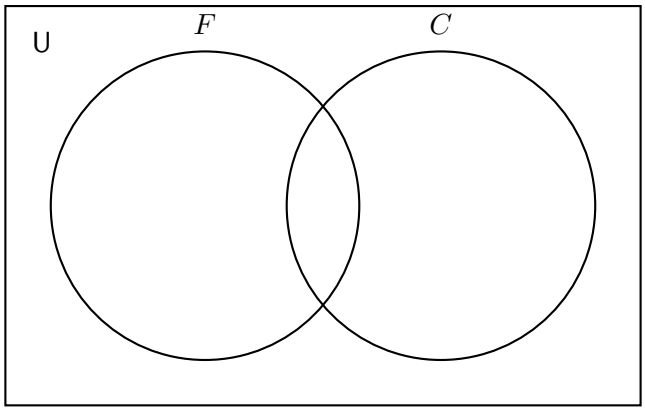
(c) Find the probability that the two counters chosen have different colours.

..... **[3]**

25 There are 40 students in a group.

- 24 students play Football (F).
- 15 students play Cricket (C).
- 8 students play both Football and Cricket.

(a) Complete the Venn diagram to show this information.



[2]

(b) A student is chosen at random.
Calculate the probability that the student plays neither Football nor Cricket.

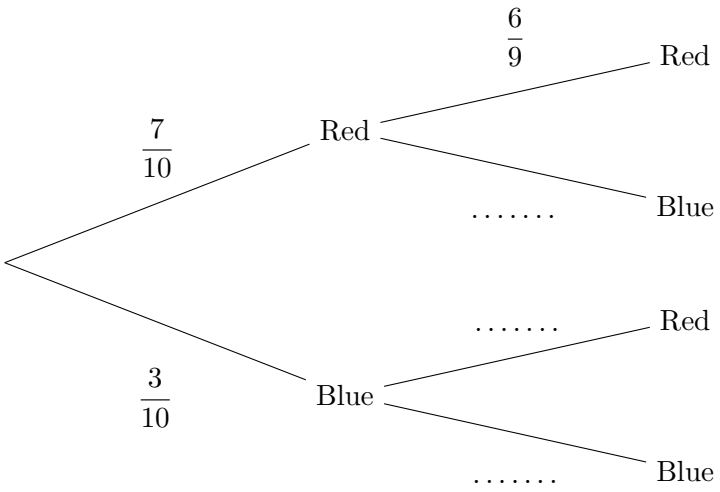
..... **[2]**

(c) Find $n(F \cup C)$.

..... **[2]**

26 A box contains 7 red counters and 3 blue counters.
Two counters are taken at random from the box, without replacement.

(a) Complete the tree diagram.



[3]

(b) Calculate the probability that both counters are red.

..... **[2]**

(c) Calculate the probability that the two counters are different colours.

..... **[2]**

27 A bag contains x green apples and 4 red apples.
Two apples are taken at random, without replacement.

(a) The probability that both apples are green is $\frac{5}{18}$.

Show that $13x^2 - 53x - 60 = 0$.

[4]

(b) Solve the equation $13x^2 - 53x - 60 = 0$ to find the value of x.

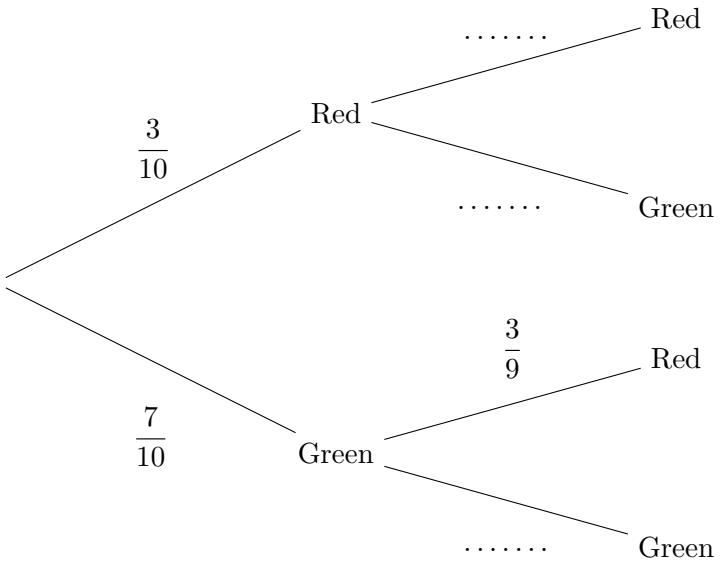
$x =$ ..... **[3]**

(c) Work out the probability that exactly one of the two apples taken is green.

..... **[3]**

28 A basket contains 7 green apples and 3 red apples.
Two apples are chosen at random from the basket, without replacement.

(a) Complete the tree diagram.



[2]

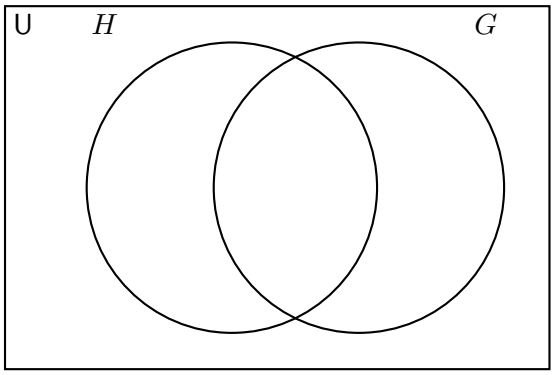
(b) Calculate the probability that both apples chosen are the same colour.

..... **[3]**

29 In a survey, 40 students are asked if they study History (H) or Geography (G).

- 23 students study History.
- 19 students study Geography.
- 7 students study neither subject.

(a) Complete the Venn diagram to show this information.



[2]

(b) A student is chosen at random from the 40 students.
Find the probability that the student studies History but not Geography.

..... **[1]**

(c) Two students are chosen at random from those who study History.
Calculate the probability that they both also study Geography.

..... **[2]**

30 A bag contains x red counters and 5 blue counters.
Two counters are taken at random from the bag without replacement.
The probability that both counters are blue is $\frac{2}{15}$.

(a) Show that $x^2 + 9x - 130 = 0$.

[3]

(b) Find the value of x.

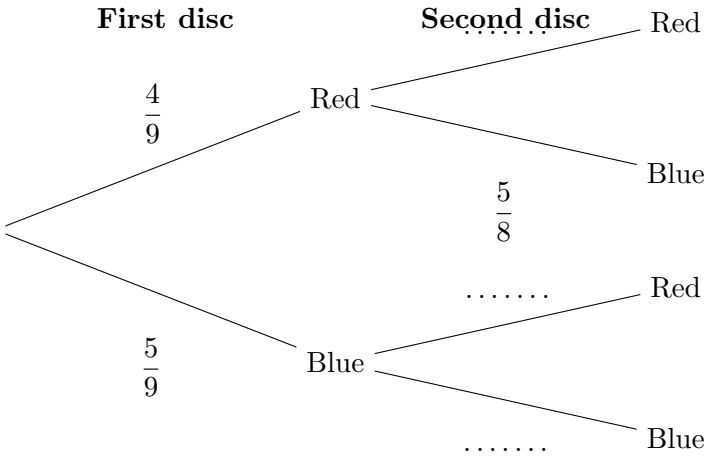
$x =$ ..... **[2]**

(c) Calculate the probability that the two counters chosen are different colours.

..... **[3]**

31 A bag contains 4 red discs and 5 blue discs.
Jamil takes one disc at random, notes its colour, and does not replace it.
He then takes a second disc at random.

(a) Complete the tree diagram.



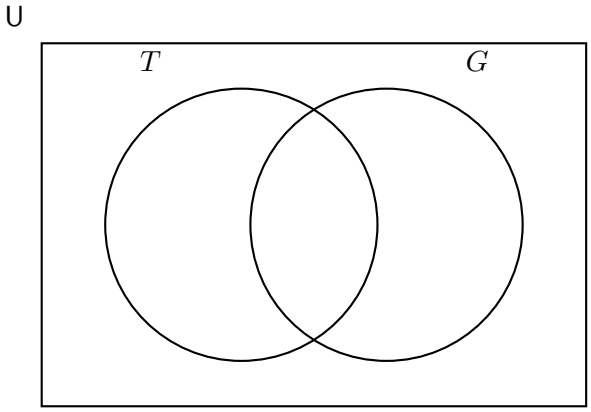
[2]

(b) Calculate the probability that Jamil takes one red disc and one blue disc.

..... **[3]**

32 In a group of 40 students, 24 play tennis (T) and 18 play golf (G).
5 students play neither tennis nor golf.

(a) Complete the Venn diagram to show this information.



[2]

(b) Two students are chosen at random from the group of 40.
Calculate the probability that both students play tennis but not golf.

..... **[3]**

(c) A student who plays golf is chosen at random.
Find the probability that they also play tennis.

..... **[2]**

33 A bag contains n sweets.
5 of the sweets are orange and the rest are lemon.
Two sweets are taken from the bag at random, without replacement.
The probability that both sweets are orange is $\frac{2}{21}$.

(a) Show that $n^2 - n - 210 = 0$.

[3]

(b) Solve the equation $n^2 - n - 210 = 0$ to find the number of sweets in the bag.

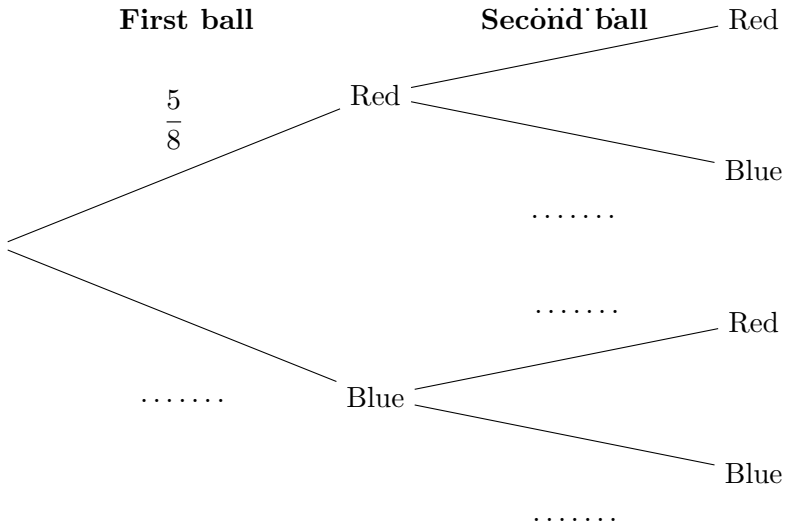
$n =$ ..... **[2]**

(c) Find the probability that the two sweets taken have different flavours.

..... **[3]**

34 A bag contains 8 balls.
There are 5 Red balls and 3 Blue balls.
A ball is taken at random from the bag, its colour is recorded, and it is replaced.
A second ball is then taken at random from the bag.

(a) Complete the tree diagram.



[2]

(b) Calculate the probability that both balls are Red.

..... **[2]**

(c) Calculate the probability that one ball is Red and one ball is Blue.

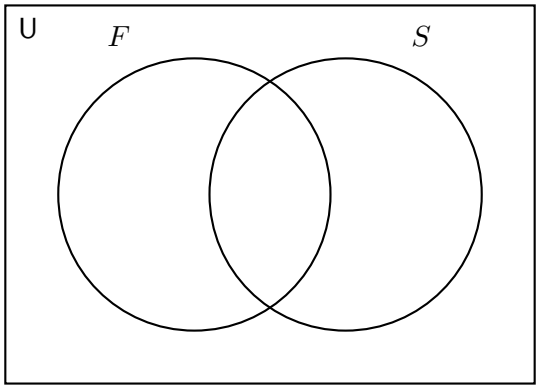
..... **[3]**

35 In a class of 30 students, 18 study French (F) and 14 study Spanish (S).
3 students study neither French nor Spanish.
Let x be the number of students who study both French and Spanish.

(a) Find the value of x.

$x =$ ..... **[2]**

(b) Complete the Venn diagram to show the number of students in each region.



[2]

(c) Two students are chosen at random from the class.
Calculate the probability that both students study French but do not study Spanish.

..... **[3]**

36 A box contains n chocolates.
There are 4 milk chocolates and the rest are dark chocolates.
Two chocolates are taken at random from the box, without replacement.
The probability that both chocolates are milk chocolate is $\frac{1}{14}$.

(a) Show that $n^2 - n - 168 = 0$.

[3]

(b) Solve the equation to find the number of dark chocolates in the box.

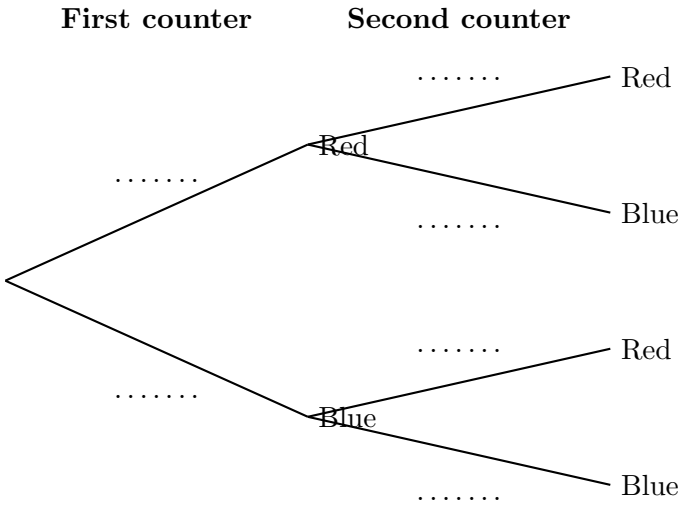
..... **[3]**

(c) Calculate the probability that the two chocolates taken are of different types.

..... **[3]**

37 A bag contains 5 red counters and 3 blue counters.
A counter is picked at random and its colour is noted.
It is then replaced in the bag.
A second counter is then picked at random.

(a) Complete the tree diagram.



[2]

(b) Work out the probability that both counters are red.

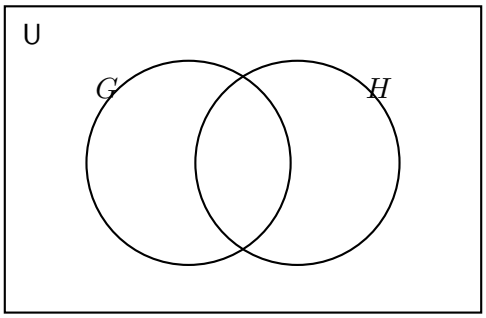
..... **[1]**

(c) Work out the probability that one counter is red and one counter is blue.

..... **[2]**

38 In a group of 40 students, 24 study Geography (G), 18 study History (H), and 6 study neither subject.

(a) Complete the Venn diagram to show the number of students in each region.



[2]

(b) A student is chosen at random from the group of 40 students.

(i) Find the probability that the student studies History but not Geography.

..... **[1]**

(ii) Find $P(G \cup H')$.

..... **[2]**

(c) Two students are chosen at random without replacement from the group of 40 students. Work out the probability that both students study Geography.

..... **[2]**

39 A box contains n sweets. 6 of the sweets are lemon flavour and the rest are orange flavour.

(a) Hannah takes two sweets at random from the box without replacement.
The probability that both sweets are lemon flavour is $\frac{1}{3}$.

(i) Show that $n^2 - n - 90 = 0$.

[3]

(ii) Solve the equation $n^2 - n - 90 = 0$ to find the total number of sweets in the box.

$n =$ ..... **[3]**

(b) Three sweets are now taken at random without replacement from the original box of 10 sweets.
Calculate the probability that at least one of the sweets taken is orange flavour.

..... **[3]**

1 **(a)** $\frac{3}{8}$ correctly placed on the first branch for Blue **[B1]**

$\frac{5}{8}$ and $\frac{3}{8}$ correctly placed on all second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $\frac{3}{8}$ correctly placed for 1st ball Blue **[B1]**

$\frac{3}{8}$, $\frac{5}{8}$, and $\frac{3}{8}$ correctly placed on all three remaining 2nd ball branches **[B1]**

(b) $\frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$\frac{30}{64}$ or $\frac{15}{32}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** First stage branches correctly labeled: $\frac{7}{12}$ for R and $\frac{5}{12}$ for B **[B1]**

Second stage branches from R correctly labeled: $\frac{6}{11}$ for R and $\frac{5}{11}$ for B **[B1]**

Second stage branches from B correctly labeled: $\frac{7}{11}$ for R and $\frac{4}{11}$ for B **[B1]**

(b) Multiplies probabilities along the R, R path: $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{42}{132}$ or $\frac{7}{22}$ **[A1]**

(c) Identifies and sums both correct combinations: $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ or $2 \times \left(\frac{35}{132}\right)$ **[M1]**

$\frac{70}{132}$ or $\frac{35}{66}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** Sets up correct equation for probability of two red balls: $\frac{x}{x+4} \times \frac{x-1}{x+3} = \frac{1}{3}$ **[M1]**

Cross-multiplies correctly to clear fractions: $3x(x-1) = (x+4)(x+3)$ **[M1]**

Expands and rearranges correctly with no errors:
$3x^2 - 3x = x^2 + 7x + 12 \implies 2x^2 - 10x - 12 = 0 \implies x^2 - 5x - 6 = 0$ **[A1]**

(b) Factorises or solves the quadratic equation: $(x-6)(x+1) = 0$ **[M1]**

6 (and explicitly or implicitly rejects $x = -1$) **[A1]**

(c) Uses $x = 6$ to determine probabilities for one red and one blue ball: $\frac{6}{10} \times \frac{4}{9}$ **[M1]**

Accounts for both (R, B) and (B, R) combinations: $2 \times \left(\frac{6}{10} \times \frac{4}{9}\right)$ or $\frac{24}{90} + \frac{24}{90}$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

5 (a) $\frac{3}{n} \times \frac{2}{n-1} = \frac{1}{15}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90 \implies n^2 - n - 90 = 0$ **[A1]**

(b) $(n - 10)(n + 9) = 0$ **[M1]**

$n = 10$ **[A1]**

(c) Number of Blue counters $= 7$ **[B1]**

$\left(\frac{3}{10} \times \frac{7}{9}\right) + \left(\frac{7}{10} \times \frac{3}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

6 (a) 6 placed correctly in the intersection **[B1]**

12 in F only, 8 in T only, and 4 outside the circles **[B1]**

(b) $\frac{6}{30}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(c) $\frac{12}{30}$ seen as the probability for the first student **[M1]**

$\frac{12}{30} \times \frac{11}{29}$ **[M1]**

$\frac{132}{870}$ or $\frac{22}{145}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) $2x + x + (x + 4) + 8 = 40$ **[M1]**

$4x + 12 = 40 \implies x = 7$ **[A1]**

(b) $\frac{32}{40}$ or $\frac{4}{5}$ **[B1]**

(c) Number of students playing Football only $= 14$ **[B1]**

$\frac{14}{40} \times \frac{13}{39}$ **[M1]**

$\frac{182}{1560}$ or $\frac{7}{60}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Places 6 outside both circles **[B1]**

Calculates intersection: $32 + 21 + 6 - 50 = 9$ and places 9 in $H \cap G$ **[B1]**

Places 23 in H only and 12 in G only **[B1]**

(b) **(i)** $\frac{9}{50}$ **[B1]**

(ii) $\frac{38}{50}$ or $\frac{19}{25}$ **[B1]**

(c) Recognises the reduced sample space and uses probability without replacement: $\frac{9}{32} \times \frac{8}{31}$ **[M1]**

$\frac{72}{992}$ or $\frac{9}{124}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** $\frac{4}{x} \times \frac{3}{x-1}$ **[M1]**

$\frac{12}{x(x-1)} = \frac{1}{11}$ or $12 \times 11 = x(x-1)$ **[M1]**

$132 = x^2 - x \implies x^2 - x - 132 = 0$ **[A1]**

(b) $(x-12)(x+11) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = 12$ **[A1]**

Number of blue discs $= 12 - 4 = 8$ **[A1]**

(c) $\frac{4}{12} \times \frac{8}{11}$ or $\frac{32}{132}$ **[M1]**

$\left(\frac{4}{12} \times \frac{8}{11}\right) \times 2$ **[M1]**

$\frac{64}{132}$ or $\frac{16}{33}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

10 **(a)** Intersection region $= 6$ **[B1]**

French only $= 11$, Spanish only $= 8$, outside $= 5$ **[B1]**

(b) **(i)** $\frac{6}{30}$ or $\frac{1}{5}$ **[B1]**

(ii) $\frac{11}{30}$ **[B1]**

(iii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** First branch $B = \frac{3}{10}$ **[B1]**

Top branch after R: $B = \frac{3}{9}$ **[B1]**

Bottom branches after B: $R = \frac{7}{9}$ and $B = \frac{2}{9}$ **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

(c) $\frac{7}{10} \times \frac{3}{9} + \frac{3}{10} \times \frac{7}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** $\frac{x}{x+4} \times \frac{x-1}{x+3}$ **[M1]**

$= \frac{x(x-1)}{(x+4)(x+3)} = \frac{x(x-1)}{x^2+7x+12}$ **[A1]**

(b) **(i)** $\frac{x(x-1)}{x^2+7x+12} = \frac{1}{3}$ **[M1]**

$3(x^2 - x) = x^2 + 7x + 12$ or $3x^2 - 3x = x^2 + 7x + 12$ **[M1]**

$2x^2 - 10x - 12 = 0 \implies x^2 - 5x - 6 = 0$ **[A1]**

(ii) $(x-6)(x+1) = 0$ **[M1]**

$x = 6$ or $x = -1$ **[A1]**

Rejects $x = -1$ to conclude there are 6 blue counters **[B1]**

[Back to Question] *[Total Marks: 8]*

13 **(a)** Number of students studying both B and C: $(18 + 15) - (30 - 4) = 7$ **[M1]**

Fully correct Venn diagram: Only $B = 11$, $B \cap C = 7$, Only $C = 8$, Neither $= 4$ **[A1]**

(b) **(i)** $\frac{8}{30}$ or $\frac{4}{15}$ **[B1]**

(ii) $\frac{26}{30}$ or $\frac{13}{15}$ **[B1]**

(iii) $\frac{7}{18}$ **[B1]**

[Back to Question] *[Total Marks: 5]*

14 **(a)** First pick: $\mathrm{P}(R) = \frac{5}{9}$ and $\mathrm{P}(Y) = \frac{4}{9}$ **[B1]**

Second pick: After R, $\mathrm{P}(R) = \frac{4}{8}$, $\mathrm{P}(Y) = \frac{4}{8}$; After Y, $\mathrm{P}(R) = \frac{5}{8}$, $\mathrm{P}(Y) = \frac{3}{8}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(YY) = \left(\frac{5}{9} \times \frac{4}{8}\right) + \left(\frac{4}{9} \times \frac{3}{8}\right)$ **[M1]**

$\frac{20}{72} + \frac{12}{72} = \frac{32}{72} = \frac{4}{9}$ **[A1]**

(c) Method 1: $1 - \mathrm{P}(RR)$ or Method 2: $\mathrm{P}(RY) + \mathrm{P}(YR) + \mathrm{P}(YY)$ **[M1]**

$1 - \left(\frac{5}{9} \times \frac{4}{8}\right) = 1 - \frac{20}{72}$ **[M1]**

$\frac{52}{72} = \frac{13}{18}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $\mathrm{P}(\text{Blue from A}) = \frac{x}{x+3}$ and $\mathrm{P}(\text{Red from A}) = \frac{3}{x+3}$ **[B1]**

If Blue transferred to B, Box B has 5 Blue and 2 Red (total 7), so $\mathrm{P}(\text{Blue from B}) = \frac{5}{7}$ **[B1]**

If Red transferred to B, Box B has 4 Blue and 3 Red (total 7), so $\mathrm{P}(\text{Blue from B}) = \frac{4}{7}$ **[B1]**

Total probability $= \left(\frac{x}{x+3} \times \frac{5}{7}\right) + \left(\frac{3}{x+3} \times \frac{4}{7}\right) = \frac{5x+12}{7(x+3)}$ **[A1]**

(b) **(i)** $\frac{5x+12}{7(x+3)} = \frac{3}{4}$ **[M1]**

$4(5x+12) = 21(x+3) \implies 20x + 48 = 21x + 63$ **[M1]**

$x = 15$ **[A1]**

(ii) $\frac{\mathrm{P}(\text{Red from A and Blue from B})}{\mathrm{P}(\text{Blue from B})} = \frac{\frac{3}{18} \times \frac{4}{7}}{\frac{3}{4}} = \frac{\frac{1}{14}}{\frac{3}{4}} = \frac{2}{21}$ **[B1]**

[Back to Question] *[Total Marks: 8]*

16 **(a)** $\frac{6}{9}$ and $\frac{3}{9}$ correctly placed on the top branches **[B1]**

$\frac{7}{9}$ and $\frac{2}{9}$ correctly placed on the bottom branches **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\left(\frac{7}{10} \times \frac{6}{9}\right) + \left(\frac{3}{10} \times \frac{2}{9}\right)$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

17 **(a)** 3 placed correctly outside the circles **[B1]**

x, $24 - x$, and $18 - x$ placed correctly in the regions **[B1]**

(b) $(24 - x) + x + (18 - x) + 3 = 40$ or equivalent **[M1]**

5 **[A1]**

(c) $\frac{19}{40} \times \frac{13}{39}$ **[M1]**

$2 \times \left(\frac{19}{40} \times \frac{13}{39}\right)$ **[M1]**

$\frac{19}{60}$ or $\frac{494}{1560}$ or 0.316... **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$90 = n^2 - n$

$n^2 - n - 90 = 0$ [A1]

(b) $(n - 10)(n + 9) = 0$ **[M1]**

10 **[A1]**

(c) $n = 10$, giving 3 Red and 7 Blue balls **[B1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

19 **(a)** Number studying both $= (18 + 15) - (30 - 5) = 8$ **[M1]**

Fully correct Venn diagram: B only $= 10$, $B \cap C = 8$, C only $= 7$, Outside $= 5$ **[A1]**

(b) **(i)** $\frac{10}{30}$ or $\frac{1}{3}$ **[B1]**

(ii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

(c) Identifies total number of Chemistry students as denominator: 15 **[M1]**

$\frac{8}{15}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

20 **(a)** First branches correct: $\mathrm{P}(R_1) = \frac{5}{8}$ and $\mathrm{P}(B_1) = \frac{3}{8}$ **[B1]**

Second branches after R_1 correct: $\mathrm{P}(R_2) = \frac{4}{7}$ and $\mathrm{P}(B_2) = \frac{3}{7}$ **[B1]**

Second branches after B_1 correct: $\mathrm{P}(R_2) = \frac{5}{7}$ and $\mathrm{P}(B_2) = \frac{2}{7}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(BB) = \left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{3}{8} \times \frac{2}{7}\right)$ **[M1]**

$\frac{26}{56}$ or $\frac{13}{28}$ (or 0.464 or 0.4642...) **[A1]**

(c) $1 - \mathrm{P}(BB) = 1 - \left(\frac{3}{8} \times \frac{2}{7}\right)$ or $\left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{5}{8} \times \frac{3}{7}\right) + \left(\frac{3}{8} \times \frac{5}{7}\right)$ **[M1]**

$\frac{50}{56}$ or $\frac{25}{28}$ (or 0.893 or 0.8928...) **[A1]**

[Back to Question] *[Total Marks: 7]*

21 (a) Sets up correct initial probability equation: $\frac{4}{n} \times \frac{3}{n-1} = \frac{2}{15}$ **[M1]**

Simplifies equation: $\frac{12}{n(n-1)} = \frac{2}{15} \implies 2n(n-1) = 180$ **[M1]**

Expands and rearranges correctly to establish $n^2 - n - 90 = 0$ with no missing algebraic steps **[A1]**

(b) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Rejects negative root $n = -9$ and states $n = 10$ **[A1]**

(c) Identifies total number of chocolates $n = 10$, dark = 4, milk = 6 **[B1]**

Calculates probability of one correct arrangement (e.g., DD M): $\frac{4}{10} \times \frac{3}{9} \times \frac{6}{8} = \frac{72}{720}$ or $\frac{1}{10}$ **[M1]**

Multiplies single selection probability by 3 (accounting for DD M, D MD, M DD) **[M1]**

$\frac{216}{720}$ or $\frac{3}{10}$ (or 0.3) **[A1]**

[Back to Question] *[Total Marks: 9]*

22 (a) $\frac{2}{5}$ correctly placed on all three Tails branches **[B1]**

$\frac{3}{5}$ correctly placed on the remaining Heads branch **[B1]**

(b) $\frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{3}{5}$ or $2 \times \left(\frac{3}{5} \times \frac{2}{5}\right)$ **[M1]**

$\frac{12}{25}$ or 0.48 **[A1]**

[Back to Question] *[Total Marks: 4]*

23 (a) 10 in the intersection **[B1]**

14 in F only, 9 in T only, and 7 outside the circles **[B1]**

(b) $\frac{10}{40}$ or $\frac{1}{4}$ **[B1]**

(c) $\frac{10}{24} \times \frac{14}{23}$ or $\frac{14}{24} \times \frac{10}{23}$ **[M1]**

$\left(\frac{10}{24} \times \frac{14}{23}\right) + \left(\frac{14}{24} \times \frac{10}{23}\right)$ **[M1]**

$\frac{35}{69}$ or $\frac{280}{552}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

24 (a) $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90$

$n^2 - n - 90 = 0$ with no errors seen [A1]

(b) $(n-10)(n+9) = 0$ leading to $n = 10$ **[M1]**

7 **[A1]**

(c) $\frac{3}{10} \times \frac{7}{9}$ or $\frac{7}{10} \times \frac{3}{9}$ **[M1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{7}{15}$ or $\frac{42}{90}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

25 (a) 8 in the intersection **[B1]**

16 in F only, 7 in C only, and 9 outside the circles **[B1]**

(b) Identifies 9 as the number of students playing neither **[M1]**

$\frac{9}{40}$ **[A1]**

(c) $16 + 8 + 7$ or $40 - 9$ **[M1]**

31 **[A1]**

[Back to Question] *[Total Marks: 6]*

26 (a) $\frac{3}{9}$ on the Red

Bluebranch[*B*1]

$\frac{7}{9}$ on the Blue

Redbranch[*B*1]

$\frac{2}{9}$ on the Blue

Bluebranch[*B*1]

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

(c) $\left(\frac{7}{10} \times \frac{3}{9}\right) + \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

27 **(a)** $\frac{x}{x+4} \times \frac{x-1}{x+3} = \frac{5}{18}$ **[M1]**

$18x(x-1) = 5(x+4)(x+3)$ **[M1]**

$18x^2 - 18x = 5(x^2 + 7x + 12)$ **[M1]**

$13x^2 - 53x - 60 = 0$ obtained with no errors **[A1]**

(b) $(13x + 12)(x - 5) = 0$ or correct use of the quadratic formula [M2] (allow [M1] for one error in formula)

$x = 5$ or $x = -\frac{12}{13}$ **[A1]**

Selects $x = 5$ as the only valid solution for the number of apples **[B1]**

(c) Total apples = 9, green = 5, red = 4. Evaluates one branch: $\frac{5}{9} \times \frac{4}{8}$ or $\frac{4}{9} \times \frac{5}{8}$ **[M1]**

$\left(\frac{5}{9} \times \frac{4}{8}\right) + \left(\frac{4}{9} \times \frac{5}{8}\right)$ or multiplies one branch by 2 **[M1]**

$\frac{40}{72}$ or $\frac{5}{9}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

28 **(a)** $\frac{6}{9}$ on the Green branch from Green **[B1]**

$\frac{2}{9}$ and $\frac{7}{9}$ on the branches from Red **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{7}{10} \times \frac{6}{9} + \frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

29 **(a)** 9 in the intersection of H and G **[B1]**

14, 10, and 7 in the correct regions **[B1]**

(b) $\frac{14}{40}$ or $\frac{7}{20}$ **[B1]**

(c) $\frac{9}{23} \times \frac{8}{22}$ **[M1]**

$\frac{72}{506}$ or $\frac{36}{253}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

30 **(a)** $\frac{5}{x+5} \times \frac{4}{x+4} = \frac{2}{15}$ **[M1]**

$20 \times 15 = 2(x+5)(x+4)$ or $300 = 2(x^2 + 9x + 20)$ **[M1]**

$150 = x^2 + 9x + 20$

$x^2 + 9x - 130 = 0$ (No errors seen) [A1]

(b) $(x+13)(x-10) = 0$ **[M1]**

$x = 10$ **[A1]**

(c) Total counters = 15 **[B1]**

$\left(\frac{10}{15} \times \frac{5}{14}\right) + \left(\frac{5}{15} \times \frac{10}{14}\right)$ **[M1]**

$\frac{100}{210}$ or $\frac{10}{21}$ **[A1]**

[Back to Question] *[Total Marks: 8]*

31 **(a)** $\frac{3}{8}$ correctly placed for top branch (Red given Red) **[B1]**

$\frac{4}{8}$ correctly placed for bottom branches (Red given Blue, and Blue given Blue) **[B1]**

(b) $\frac{4}{9} \times \frac{5}{8}$ or $\frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{4}{9} \times \frac{5}{8} + \frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{5}{9}$ or equivalent (e.g. $\frac{40}{72}$) **[A1]**

[Back to Question] *[Total Marks: 5]*

32 **(a)** 7 correctly placed in the intersection of T and G **[B1]**

17, 11, and 5 correctly placed in the remaining regions **[B1]**

(b) Identifies 17 students play tennis but not golf **[B1]**

$\frac{17}{40} \times \frac{16}{39}$ **[M1]**

$\frac{34}{195}$ or 0.174 (0.1743...) **[A1]**

(c) Denominator 18 seen **[M1]**

$\frac{7}{18}$ or 0.389 (0.3888...) **[A1]**

[Back to Question] *[Total Marks: 7]*

33 (a) $\frac{5}{n} \times \frac{4}{n-1} = \frac{2}{21}$ [M2]

(Award M1 for $\frac{5}{n} \times \frac{4}{n}$ or $\frac{5}{n} + \frac{4}{n-1}$)

$2(n^2 - n) = 420$ or $n(n-1) = 210$ **[M1]**

$n^2 - n - 210 = 0$ reached with no errors seen **[A1]**

(b) $(n-15)(n+14) = 0$ or correct substitution into quadratic formula **[M1]**

$n = 15$ **[A1]**

(c) $\frac{5}{15} \times \frac{10}{14}$ or $\frac{10}{15} \times \frac{5}{14}$ seen **[M1]**

Multiplies their probability by 2 (i.e. $2 \times \frac{50}{210}$) **[M1]**

$\frac{10}{21}$ or 0.476 (0.4761...) **[A1]**

[Back to Question] *[Total Marks: 8]*

34 (a) $\frac{3}{8}$ placed correctly on the lower first branch **[B1]**

$\frac{5}{8}$, $\frac{3}{8}$ and $\frac{5}{8}$, $\frac{3}{8}$ placed correctly on the second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

(c) $\frac{5}{8} \times \frac{3}{8}$ or $\frac{15}{64}$ **[M1]**

$2 \times \left(\frac{5}{8} \times \frac{3}{8}\right)$ **[M1]**

$\frac{15}{32}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

35 (a) $18 + 14 + 3 - x = 30$ or $35 - x = 30$ **[M1]**

5 **[A1]**

(b) 5 placed correctly in the intersection **[B1]**

13 in F only, 9 in S only, and 3 outside the circles **[B1]**

(c) Identifies number of students studying only French is 13 **[M1]**

$\frac{13}{30} \times \frac{12}{29}$ **[M1]**

$\frac{156}{870}$ or $\frac{26}{145}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

36 (a) $\frac{4}{n} \times \frac{3}{n-1} = \frac{1}{14}$ **[M1]**

$\frac{12}{n^2 - n} = \frac{1}{14}$ **[M1]**

$168 = n^2 - n$ leading to $n^2 - n - 168 = 0$ with no errors seen **[A1]**

(b) $(n - 14)(n + 12) = 0$ or correct use of quadratic formula **[M1]**

$n = 14$ (rejecting $n = -12$) **[A1]**

Number of dark chocolates $= 14 - 4 = 10$ **[A1]**

(c) $\frac{4}{14} \times \frac{10}{13}$ or $\frac{10}{14} \times \frac{4}{13}$ **[M1]**

$2 \times \left(\frac{4}{14} \times \frac{10}{13}\right)$ **[M1]**

$\frac{80}{182}$ or $\frac{40}{91}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

37 (a) First counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ **[B1]**

Second counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ on both sets of branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$= \frac{30}{64}$ (or $\frac{15}{32}$) **[A1]**

[Back to Question] *[Total Marks: 5]*

38 (a) Intersection $n(G \cap H) = 8$ correctly placed **[B1]**

Remaining regions: $n(G \cap H') = 16$, $n(H \cap G') = 10$, outside region $= 6$ **[B1]**

(b) (i) $\frac{10}{40}$ (or $\frac{1}{4}$ or 0.25) **[B1]**

(ii) Identifies regions $16 + 8 + 6 = 30$ or calculates $1 - \frac{10}{40}$ **[M1]**

$\frac{30}{40}$ (or $\frac{3}{4}$ or 0.75) **[A1]**

(c) $\frac{24}{40} \times \frac{23}{39}$ **[M1]**

$= \frac{552}{1560}$ (or $\frac{23}{65}$ or 0.354) **[A1]**

[Back to Question] *[Total Marks: 7]*

39 **(a)** **(i)** Sets up equation $\frac{6}{n} \times \frac{5}{n-1} = \frac{1}{3}$ **[M1]**

Simplifies to $\frac{30}{n(n-1)} = \frac{1}{3}$ or $30 \times 3 = n(n-1)$ **[M1]**

$n^2 - n = 90$

$n^2 - n - 90 = 0$ with no steps omitted [A1]

(ii) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Obtains $n = 10$ and $n = -9$ **[A1]**

Rejects negative root and states total number of sweets = 10 **[B1]**

(b) Calculates $P(\text{all 3 lemon}) = \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8}$ **[M1]**

Identifies $P(\text{at least 1 orange}) = 1 - P(\text{all 3 lemon})$ **[M1]**

$= 1 - \frac{120}{720} = 1 - \frac{1}{6} = \frac{5}{6}$ (or 0.833) **[A1]**

[Back to Question] *[Total Marks: 9]*